EXAMPLE NOTE DOCUMENT

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Abstract. This short note showcases the note template: theorem-like environments, theorem-numbered lists, ordinary enumerated lists, displayed equations, commutative diagrams, citations, URLs, and cross-references.

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1. Objects and diagrams

Definition 1.1. A *good object* is an object X together with a morphism

$$\eta_X \colon X \to F(X)$$

satisfying the following conditions.

(a) The assignment $X \mapsto \eta_X$ is functorial in X.
(b) The morphism η_X is compatible with base change.

The numbering in Items 1.1(a) and 1.1(b) is derived from the surrounding theorem-like environment Theorem 1.1. This is useful for statements with several numbered conditions.

Example 1.2. A basic diagrammatic situation is a commutative square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[g]{} & Y. \end{array}$$

The command `tikz-cd` is available in this template for diagrams of this kind.

Lemma 1.3. *In the situation of Theorem 1.2, suppose that the square is cartesian. Then the compatibility condition in Item 1.1(b) can be expressed by the identity*

$$g^* f_* \simeq f'_* g'^*. \tag{1.1}$$

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Proof. This is only a template proof. The displayed formula illustrates a typical mathematical display, while the reference to Item 1.1(b) illustrates a theorem-numbered item reference. □

2. References and ordinary lists

The template uses biblatex. For example, one may cite Hartshorne's textbook [Har77] or the Stacks Project [SP].

Ordinary lists remain ordinary. For instance:

(a) This is a first ordinary item.
(b) This is a second ordinary item.

Thus Item (b) is an ordinary item reference, whereas Item 1.1(b) is a theorem-numbered item reference. ⚠ A to-do note here.

Proposition 2.1. *Let X be a good object. If the square in Theorem 1.2 is cartesian, then X' inherits the corresponding compatibility property.*

Proof. The assertion follows formally from Theorem 1.3 and Item 1.1(b). **Bob:** Maybe we should add more details here. □

Alice: We should add a remark here.

Remark 2.2. Hyperlinks can be written using `\href`. For example, the template repository could link to the tikz-cd package page. URLs inherit the hyperlink colour configured in the common preamble.

Bob: I've added the remark above.

References

[Har77] Robin Hartshorne. *Algebraic Geometry*. Vol. 52. Graduate Texts in Mathematics. Springer, 1977.

[SP] The Stacks Project Authors. *The Stacks Project*. URL: https://stacks.math.columbia.edu.

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